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COMMERCIAL ITEM DESCRIPTION

MILLING MACHINE, HORIZONTAL SPINDLE, KNEE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers knee horizontal-spindle knee milling machines intended for use in conventional milling operations.

2. CLASSIFICATION. The milling machines shall be of the following types, classes, and sizes. The type, class, and size to be furnished shall be as specified (see 7.2(b)).

Type I - Plain worktable
Type II - Universal worktable

Class 1 - 5-horsepower spindle drive
Class 2 - 7.5-horsepower spindle drive
Class 3 - 10-horsepower spindle drive
Class 4 - 20-horsepower spindle drive

Size - Sizes are shown in table I

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Supply Center Richmond, ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3417

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TABLE I. Horizontal-spindle milling machine sizes.

Size	Class	Minimum machine travel, inches		
		Longitudinal	Cross-slide	Vertical
1	1	22	12	15
2		29	9	16
3		37	13	18
4	2	29	12	17
5		35	14	17
6		41	13	18
7	3	31	10	15
8		37	12	14
9		45	15	19
10	4	52	14	24

3. SALIENT CHARACTERISTICS

3.1 General requirements. The milling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The milling machine shall include, as a minimum, the following components:

3.2.1 Column and base. The milling machine column and base shall support the other elements of the machine, provide stability for the worktable, minimize adverse vibration, and provide a means for leveling and anchoring the machine.

3.2.2 Knee. The knee shall support the worktable and provide for locking the worktable within its travel range on the column. The knee adjusting crank lever or handwheel shall automatically release when the power feed or rapid traverse is engaged or power feed to the vertical or cross axes shall be disabled when the crank lever or handwheel is engaged for that axis.

3.2.3 Worktable. Unless otherwise specified (see 7.2(c)), the worktable length, width, and workpiece weight capacity shall be that considered as standard by the manufacturer. The number and size of the T-slots shall be as specified (see 7.2(d)). The configuration of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) B5.1. On type II machines, the worktable shall swivel horizontally at least 45 degrees clockwise and 45 degrees counterclockwise from center. The swiveling worktable shall have a position scale graduated in increments not greater than 1 degree to indicate its angular setting.

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3.2.4 Spindle assembly. Unless otherwise specified (see 7.2(e)), the spindle shall have 12 speeds over a speed range from 40 to 1600 revolutions per minute. The taper size of the spindle shall be as specified (see 7.2(f)) and shall be in accordance with ANSI B5.18.

3.2.5 Digital readout. Unless otherwise specified (see 7.2(g)), a digital readout shall be provided for positional readout of all axes. The resolution and accuracy shall be within ±0.0005 inch. The repeatability shall be within ±0.0002 inch measured over the full travel on each axis. The digital readout shall indicate axis location in both English and metric units.

3.2.6 Vertical-spindle milling attachment. When specified (see 7.2(h)), a removable vertical-spindle milling attachment shall be furnished and bolted to the column, and driven by the horizontal spindle. The milling attachment shall swivel not less than ±90 degrees in a vertical plane parallel to the longitudinal worktable motion. If required, the milling attachment shall also swivel from a vertical to a horizontal plane parallel to the cross-slide motion as specified (see 7.2(i)).

3.2.7 Universal milling attachment. When specified (see 7.2(j)), a removable universal milling attachment shall be furnished, bolted to the column, and driven by the horizontal spindle. The milling attachment shall swivel not less than 180 degrees in a vertical plane parallel to the longitudinal worktable motion and shall swivel not less than 180 degrees in a horizontal plane.

3.2.8 Universal dividing head. When specified (see 7.2(k)), a universal dividing head shall be furnished and shall rotate and index workpieces in conjunction with longitudinal movement of the worktable. The index plates shall be those normally furnished with the manufacturer's standard model dividing head. When specified (see 7.2(l)), additional index plates shall be furnished to provide for dividing a single spindle rotation into any integral number of equal divisions from 2 to 100 when using the plain indexing method. When specified (see 7.2(m)), a complete driving mechanism with enclosed gears shall be provided to drive the dividing head by power from the worktable feed screw. The dividing head shall be complete with centers, steady rest, tailstock, and three-jaw universal chuck with reversible jaws.

3.2.9 Rotary table. If required, a rotary table with a peripheral scale graduated in 1- degree increments shall be furnished and shall include a handwheel, T-slots, T-bolts, nuts, washers, and locking clamp for the rotary motor as specified (see 7.2(n)). The table diameter and workpiece weight capacity shall be as specified (see 7.2(o)). When specified (see 7.2(p)), the table shall be power driven, and two trip dogs shall be furnished. If required, an indexing attachment with indexing plates shall be furnished as specified (see 7.2(q)).

3.2.10 Controls. As a minimum, the milling machine shall be equipped with the following controls:

3.2.10.1 Manual feed controls. The milling machine shall be equipped with crank levers or handwheels to elevate the knee and move the worktable. The crank levers or handwheels shall be equipped with micrometer scales graduated in 0.001-inch increments.

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3.2.10.2 Power feed controls. The milling machine shall be equipped with a feed rate selector that shall control the power feed rate for worktable and knee in inches per minute. The power feed shall be engaged by individual controls on the worktable and knee. Trip dogs at axis travel limits shall be provided to automatically disengage the power feed to prevent accidental damage to the feed mechanism. Adjustable power feed trip dogs shall be provided to preset the point at which the feed is to be disengaged. Unless otherwise specified (see 7.2(r)), the machine shall be capable of 12 feed rates over a range from 1.5 to 25 inches per minute on the longitudinal and cross-slide axes, and from 1 to 7 inches per minute on the vertical axis.

3.2.10.3 Rapid traverse. The milling machine shall be equipped with rapid traverse controls of the knee, saddle, and worktable to move the workpiece up to the cutter or clear of the overarm. When engaged, the rapid traverse shall override the feed rate of the worktable, saddle, or knee in the direction of the feed. Unless otherwise specified (see 7.2(s)), the longitudinal, cross-slide, and vertical axis rapid traverse rates shall be not less than 63 inches per minute, 63 inches per minute, and 20 inches per minute, respectively.

3.2.10.4 Spindle controls. The milling machine shall be equipped with spindle controls including the main motor forward and reverse switch, clutch, and spindle speed controls.

3.2.10.5 Locks. The milling machine shall be equipped with locks on the worktable, saddle, and knee to permit the components to be locked in place during machining operations.

3.2.10.6 Automatic cycle worktable control. When specified (see 7.2(t)), the milling machine shall have automatic cycle worktable control of the following functions: power feed, rapid traverse, automatic reverse, automatic knee retraction, and stop.

3.2.11 Coolant system. Unless otherwise specified (see 7.2(u)), the cutting tool coolant system shall be the flood type and shall provide a means for directing and controlling the rate of coolant flow. The coolant returning to the central reservoir shall be filtered through a 1000-micron filter.

3.2.12 Electrical system. Unless otherwise specified (see 7.2(v)), the electrical system shall conform to the ANSI/National Fire Protection Association (NFPA) 79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(w)).

3.2.13 Electric motor. The spindle drive motor shall have a 30-minute duty cycle horsepower rating, shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. Unless otherwise specified (see 7.2(x)), the motors shall have open dripproof enclosures.

3.2.14 Optional accessories. Optional accessories shall be furnished as specified (see 7.2(y)).

3.3 Performance. Unless otherwise specified (see 7.2(z)), the milling machine shall meet the alignment requirements shown in table II when tested in accordance National Aerospace Standard 985.

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TABLE II. Alignment requirements.

NAS 985 test number	Test parameter
4.3.2.1	Flatness - work-mounting surface - Maximum deviation: 0.0008 inch per foot - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for either length or width
4.3.3.1	Rise and fall of work-mounting surface -Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for either length or width
4.3.4.1	T-slots parallel with longitudinal movement - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for the total T-slot length
4.3.5.1	Transverse movement square with center T-slot - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for total travel up to 0.003 inch
4.3.6.1	Center T-slot square with spindle - Maximum deviation shall not exceed 0.001 inch in 18-inch sweep - Compensate for any error indicated in test section 4.3.4.1
4.3.9.1	Transverse movement square with longitudinal movement - Maximum deviation in any 12 inches: 0.0008 inch - Maximum overall accumulated error shall not exceed 0.0008 inch plus 0.00015 inch per foot for total travel up to 0.003 inch
4.3.10.1	Spindle runout - Radial runout: 0.0003 TIR - Axial (face) runout: 0.0002 TIR - 1.25 inches from face: 0.0003 TIR - 12 inches from face: 0.001 TIR
4.3.13.1	Alignment of spindle with arbor support - Runout shall not exceed 0.001 TIR with arbor support clamped
4.3.14.1	Spindle parallel with overarm - Maximum deviation shall not exceed 0.002 TIR overall
4.3.15.1	Saddle transverse movement parallel with spindle - Maximum deviation shall not exceed 0.00075 inch per foot
4.3.17.1	Spindle parallel with work-mounting surface - Maximum deviation shall not exceed 0.001 inch per foot with worktable in unclamped position
4.3.18.1	Work-mounting surface square with vertical knee movement - Maximum deviation shall not exceed 0.005 inch per foot
4.3.21.1	Deflection of worktable under load - Worktable shall support 350 pounds per square foot on overhanging portion with a maximum deflection of 0.015 inch
4.3.22.1	Slide displacement due to clamping - Movement of knee assembly when clamp is locked shall not exceed 0.001 inch

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3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(aa)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.8. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(bb)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(cc)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(dd)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(ee)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

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5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The milling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(ff)), the milling machine shall be tested in accordance with sections 5.3.1 through 5.3.4.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(gg)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(hh)). Unless otherwise specified (see 7.2(ii)), the milling machine shall be tested in accordance with the environmental conditions specified in the American Society of Mechanical Engineers (ASME) B5.54.

5.3.2 Operational test. The milling machine shall be operated in accordance with the manufacturer's standard operating procedures for warm-up and operation. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. The alignment and cutting performance tests shall be conducted in accordance with NAS 985 and NAS 979, respectively.

5.3.3.1 Alignment tests. The alignment test results shall meet the requirements shown in table II.

5.3.3.2 Maximum torque cutting test. The machine shall meet the performance requirements of NAS 979 paragraph 4.3.3.1.1.

5.3.3.3 Maximum rated horsepower cutting tests. The machine shall meet the performance requirements of NAS 979 paragraph 4.3.3.2.2. The test shall be performed using a 4-inch width of cut, a 8-inch-per-minute feed rate, a 250-revolutions-per-minute spindle speed, a 0.0053-inch-per-tooth chip load, and a type IX cutter shown in table I of NAS 979. The depth of cut for class 1 machines shall be 0.094 inch, for class 2 machines 0.141 inch, for class 3 machines 0.188 inch, and for class 4 machines 0.375 inch.

5.3.3.4 Maximum feed rate cutting test. The machine shall meet the performance requirements of NAS 979 paragraph 4.3.3.3.1. The test shall be performed using an 11-inch width of cut, a 25-inch-per-minute feed rate, a 1200-revolutions-per-minute spindle speed, a 0.0052-inch-per-tooth chip load, and a type III cutter shown in table I of NAS 979. The depth of cut for class 1 machines shall be 0.070 inch, for class 2 machines 0.110 inch, for class 3 machines 0.145 inch, and for class 4 machines 0.290 inch.

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5.3.3.5 Flatness and mismatch cutting test. The machine shall meet the performance requirements of NAS 979 paragraph 4.3.3.12.2.
5.3.4 Optional tests. Optional tests shall be conducted as specified (see 7.2(jj)).

5.4 Acceptance. Unless otherwise specified (see 7.2(kk)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared to the performance requirements stated in sections 5.3.3 through 5.3.4 and as shown in table II. Failure of the machine to meet these performance requirements shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.2(ll)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Source of documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

Aerospace Industries Association of America (AIAA)

NAS 979	Uniform Cutting Tests - NAS Series Metal Cutting Equipment Specifications
NAS 985	Uniform Alignment Test - NAS 900 Series Equipment Specifications

Applications for copies should be sent to the Aerospace Industries Association of America, 1725 De Sales Street NW, Washington, DC 20036

American National Standards Institute (ANSI)

ANSI B5.1	T-slots, Their Bolts, Nuts, Tongues, and Cutters
ANSI B5.18	Spindle Noses and Arbors for Milling Machines
ANSI B11.8	Machine Tools – Drilling, Milling, and Boring Machines – Safety Requirements for Construction, Care, and Use and Boring Machines and Machine Tools.

Applications for copies should be sent to the American National Standards Institute, 11 West

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42nd Street, New York, NY 10036.
American Society of Mechanical Engineers (ASME)

ASME B5.54 Methods for Performance Evaluation of Computer Numerically Controlled Machining Centers

Applications for copies should be sent to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

National Electrical Manufacturers Association (NEMA)

NEMA MG-1 Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

National Fire Protection Association (NFPA)

ANSI/NFPA-79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.

Occupational Safety and Health (OSHA)

29 CFR PART 1910

Copies of OSHA documents may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type, class, and size required (see 2).
c. Worktable length, width, and workpiece weight capacity, if different (see 3.2.3).
d. Number and size of T-slots, as specified (see 3.2.3).
e. Spindle speed range and number of spindle speeds, if different (see 3.2.4).
f. Spindle taper, as specified (see 3.2.4).
g. Digital readout, if different (see 3.2.5).
h. Vertical-spindle milling attachment, if required (see 3.2.6).
i. Vertical-spindle milling attachment swivel motion, as specified (see 3.2.6).
j. Universal milling attachment, if required (see 3.2.7).
k. Universal dividing head, if required (see 3.2.8).
l. Additional index plates, if required (see 3.2.8).
m. Driving mechanism, if required (see 3.2.8).
n. Rotary table, as specified (see 3.2.9).
o. Rotary table diameter and workpiece weight capacity, as specified (see 3.2.9).

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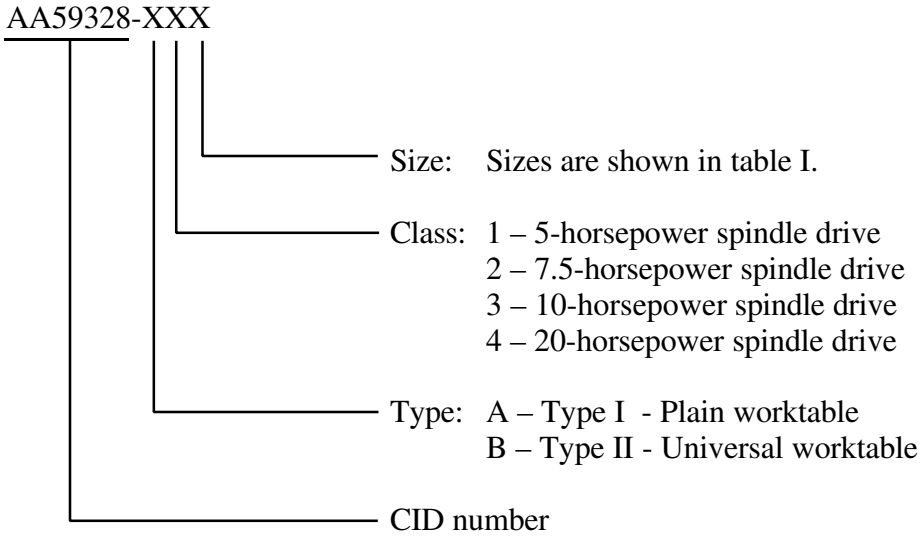
p. Powered rotary table and two trip dogs, if required (see 3.2.9).
q. Rotary table indexing attachment, as specified (see 3.2.9).
r. Feeds rates, if different (see 3.2.10.2).
s. Rapid traverse rates, if different (see 3.2.10.3).
t. Automatic table cycle control, if required (see 3.2.10.6).
u. Coolant system, if different (see 3.2.11).
v. Electrical system industrial standard, if different (see 3.2.12).
w. Electrical system voltage, as specified (see 3.2.12).
x. Motor enclosures, if different (see 3.2.13).
y. Optional accessories, as specified (see 3.2.14).
z. Performance, if different (see 3.3).
aa. Machine dimensions and weight, as specified (see 3.4).
bb. System of units required (see 3.6).
cc. Dual system of units, if required (see 3.6).
dd. Nameplate, if different (see 3.7).
ee. Lubrication plate or chart, if required (see 3.8).
ff. Testing, if different (see 5.3).
gg. Test responsibility, if different (see 5.3.1).
hh. Certification of calibration, as specified (see 5.3.1).
ii. Test environmental conditions, if different (see 5.3.1).
jj. Optional tests, as specified (see 5.3.4).
kk. Acceptance, if different (see 5.4).
ll. Packaging requirements, if different (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



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MILITARY INTERESTS:

Custodians
Air Force - 99
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Reviewers
Air Force - 84
Navy - MC

PREPARING ACTIVITY
DLA - GS

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